Adil Kaan Akan

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Education

Koc University **Istanbul, TR**
Ph.D. in Computer Science, GPA: 3.96 2022–2025
Research Focus: Object-centric learning, Generative models, Image synthesis
– Published at **ICLR 2025** on slot-guided adaptation of pre-trained diffusion models
– Extended this work to video; currently under review at **IEEE TPAMI**
Thesis: Learning Object-Centric Representations Based on Slots in Real World Scenarios

Koc University **Istanbul, TR**
M.Sc. in Computer Science, GPA: 3.96 2020–2022
Academic Excellence Award recipient
Research Focus: Stochastic video prediction, Future instance segmentation, Trajectory prediction
– Published two papers at **ICCV** on video prediction and trajectory forecasting
– Published at **ECCV** on future instance segmentation for autonomous driving
– Published two preprints on video prediction and trajectory forecasting
Thesis: Stochastic Future Prediction in Real World Driving Scenarios

Middle East Technical University **Ankara, TR**
B.Sc. in Computer Engineering, GPA: 3.83 (Ranked $5^{th}/249$) 2015–2020
Research Focus: Adversarial image generation
– Published a journal paper at **SIVP** on minimal perturbation adversarial image generation
– Published at **ICIP** on adversarial image generation

Experience

fal.ai **Remote**
Research Scientist Sep 2025–Present
- Conducting applied research on large-scale diffusion models for **image and video generation**, with a focus on production quality, controllability, and efficiency
- Exploring **RLHF / preference optimization** techniques for diffusion models to improve alignment and user control
- Distilled **FLUX.2 [dev]** into an open-weights model that **outperforms the base model**: **FLUX.2 [dev] Turbo**

Codeway Digital **Istanbul, TR**
AI Research Scientist Jan 2023–Sep 2025
- Drove research and experimentation on large-scale generative models for **Retake AI**, a personalized face and photo editor powered by diffusion models
- Built and optimized image-based generative pipelines (e.g., Stable Diffusion), enhancing visual quality and efficiency
- Designed Codeway's first proprietary generative model: adapted base diffusion models for scalable personalization and integrated with production systems
- Implemented a lightweight compression pipeline cutting model storage and infrastructure costs by over 95% without quality loss

Kuartis Technology and Consulting **Ankara, TR**
Computer Vision Engineer Feb 2020–Aug 2020
- Focused on real-time object detection for self-driving cars
- Deployed real-time object detectors and trackers on NVIDIA Drive AGX platform

ImageLab, METU **Ankara, TR**
Undergraduate Researcher Jun 2018–Aug 2020
- Brain decoding with fMRI data using machine and deep learning
- Visualization of object detectors and their receptive fields
- Adversarial image generation

Computer Vision Lab, ETH Zürich **Zurich, CH**
Research Intern at Computer-assisted Applications in Medicine group (CAiM) Jun 2019–Sep 2019

- Worked on development of a Biomedical Data Analysis tool
- Analyzed high dimensional biomedical data with image processing techniques
- Designed a UI where users can use computer vision techniques without any prior knowledge

Publications

A. Mammadov, O. Kara, K. Oktay, I. Azangulov, **A. K. Akan**, H. Chung, J.M. Rehg, Y. W. Teh, "Exact Posterior Score Estimation for Solving Linear Inverse Problems", Arxiv preprint, arXiv:2606.17048, 2026

H. Yesiltepe, J. Hu, T. Meral, **A. K. Akan**, K. Oktay, H. Eldardiry, P. Yanardag, "VideoMLA: Low-Rank Latent KV Cache for Minute-Scale Autoregressive Video Diffusion", Arxiv preprint, arXiv:2605.30351, 2026

T. Meral, K. Oktay, H. Yesiltepe, **A. K. Akan**, P. Yanardag, "Aligning Latent Geometry for Spherical Flow Matching in Image Generation", Arxiv preprint, arXiv:2605.15193, 2026

H. Yesiltepe, T. Meral, **A. K. Akan**, K. Oktay, P. Yanardag, "Infinity-RoPE: Action-Controllable Infinite Video Generation Emerges From Autoregressive Self-Rollout", Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

A. K. Akan, Y. Yemez, "Compositional Video Synthesis by Temporal Object-Centric Learning", Arxiv preprint, arXiv:2507.20855, 2025 (Submitted to IEEE TPAMI).

A. K. Akan, Y. Yemez, "Slot-Guided Adaptation of Pre-trained Diffusion Models for Object-Centric Learning and Compositional Generation", International Conference on Learning Representations (ICLR), 2025.

Gorkay Aydemir, **A. K. Akan**, F. Guney, "ADAPT: Efficient Multi-Agent Trajectory Prediction with Adaptation", Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2023.

Gorkay Aydemir, **A. K. Akan**, F. Guney, "Trajectory Forecasting on Temporal Graphs", Arxiv preprint, arXiv:2203.00255, 2022.

A. K. Akan, F. Guney, "StretchBEV: Stretching Future Instance Prediction Spatially and Temporally", In European Conference on Computer Vision (ECCV), 2022.

A. K. Akan, S. Safadoust, E. Erdem, A. Erdem, F. Guney, "Stochastic Video Prediction with Structure and Motion", Arxiv preprint, arXiv:2203.10528, 2022.

A. K. Akan, E. Erdem, A. Erdem, F. Guney, "SLAMP: Stochastic Latent Appearance and Motion Prediction", Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2021.

A. K. Akan, E. Akbas, F. T. Y. Vural, "Just Noticeable Difference for Machine Perception and Generation of Regularized Adversarial Images with Minimal Perturbation", Signal, Image and Video Processing (SIVP), 2021.

A. K. Akan, M. A. Genc, F. T. Y. Vural, "Just Noticeable Difference for Machines to Generate Adversarial Images", IEEE International Conference on Image Processing (ICIP), 2020.

A. K. Akan, B. B. Kivilcim, E. Akbas, S. D. Newman, F. T. Y. Vural, "Modeling and Decoding Complex Problem Solving Process by Artificial Neural Networks" in IEEE Signal Processing and Communications Applications Conference (SIU), 2019.

Research Interests

- Image and Video Generative models
- Object-centric Learning

Teaching

- **Reviewer:** NeurIPS2026, CVPR2026, ICLR 2026, CVPR 2025, ICLR 2025, ECCV 2024, CVPR 2024, ICCV 2023
- **Teaching:** Teaching assistant at Introduction to Computer Science, Software Engineering, Computer Engineering Design Course

Skills

Languages: Python, C/C++, Julia
Libraries and Frameworks: PyTorch, Huggingface, Diffusers, Tensorflow/Keras, Scikit-Learn
Tools: Matlab, ImageJ, HDF5, LaTeX